Retail Electricity Prices

- There are many fixed and variable costs associated with power systems, which ultimately contribute to determining retail electricity prices.
- The major *variable* operating cost is associated with generation, primarily due to fuel costs:
  - Roughly half of retail costs.

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Aside: Levelized Cost of Generation, operating costs plus paying off capital.

Technology	$/MWh (2007 Dollars) (IOU)
Advanced Nuclear	104
Wind – Class 5	67
Solar – Photovoltaic	686
Solar – Concentrating	434
Solar – Parabolic Trough	281
Ocean Wave (Pilot)	838
Small Scale Hydro	118
Geothermal	63

Keep in mind these numbers involve, assumptions that can drastically affect the value, and that many technology costs are site dependent.

Source: California Energy Commission:
http://energyalmanac.ca.gov/electricity/levelized_costs.html

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Power System Economic Operation

- Power system loads are cyclical.
- Therefore the installed generation capacity is usually much greater than the current load.
- This means that there are typically many ways we could meet the current load.
- Since different states have different mixes of generation, we will consider how generally to minimize the variable operating costs given an arbitrary, specified portfolio of generators.

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Power System Economic Operation

- Fuel cost to generate a MWh can vary widely from technology to technology.
- For some types of units, such as hydro, "fuel" costs are zero but the limit on total available water gives it an implicit value.
- For thermal units it is much easier to characterize costs.
- We will focus on minimizing the variable operating costs (primarily fuel costs) to meet demand.

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Power System Economic Operation

- Different generation technologies vary in the:
  - capital costs necessary to build the generator
  - fuel costs to actually produce electric power
- For example:
  - nuclear and hydro have high capital costs and low operating costs.
  - Natural gas generators have low capital costs, and higher operating costs.

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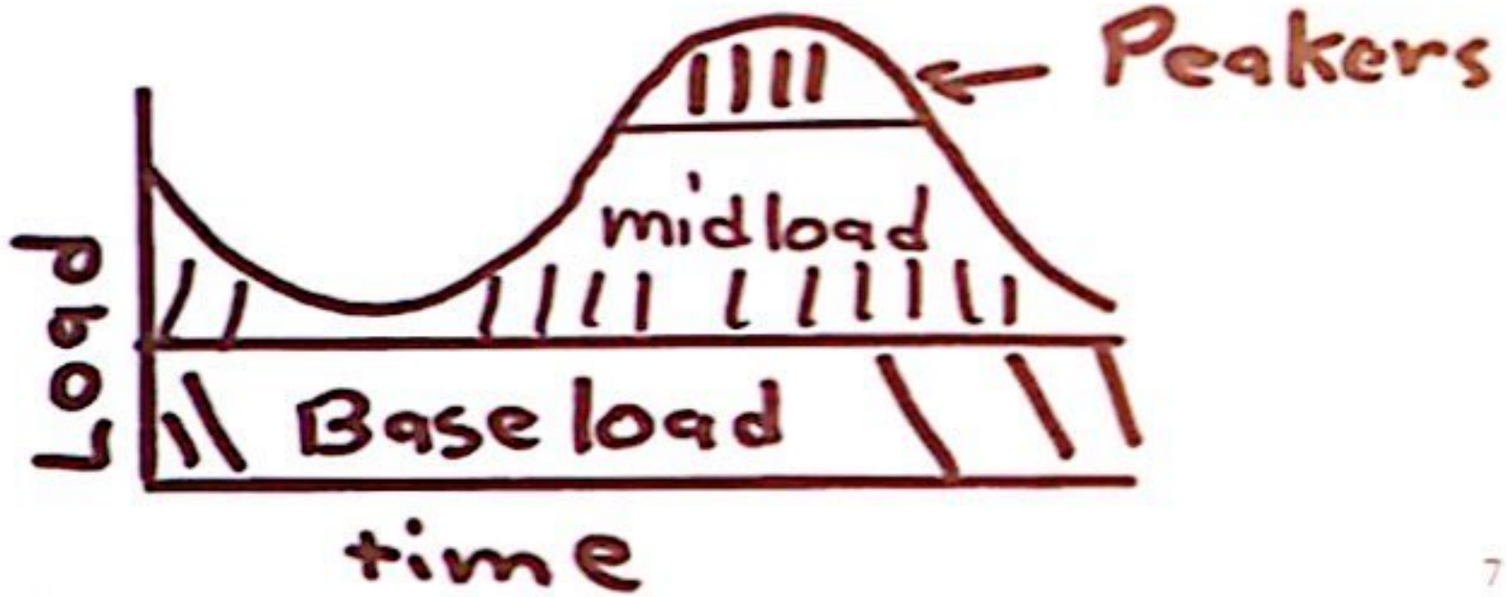
Thermal versus Hydro Generation

- The two main types of generating units are thermal and hydro, with wind and solar rapidly growing.
- For hydro the fuel (water) is free but there may be many constraints on operation:
  - fixed amounts of water available,
  - reservoir levels must be managed and coordinated,
  - downstream flow rates for fish and navigation.
- Hydro optimization is typically longer term (many months or years).
- We will concentrate on thermal units and some wind, looking at short-term optimization.

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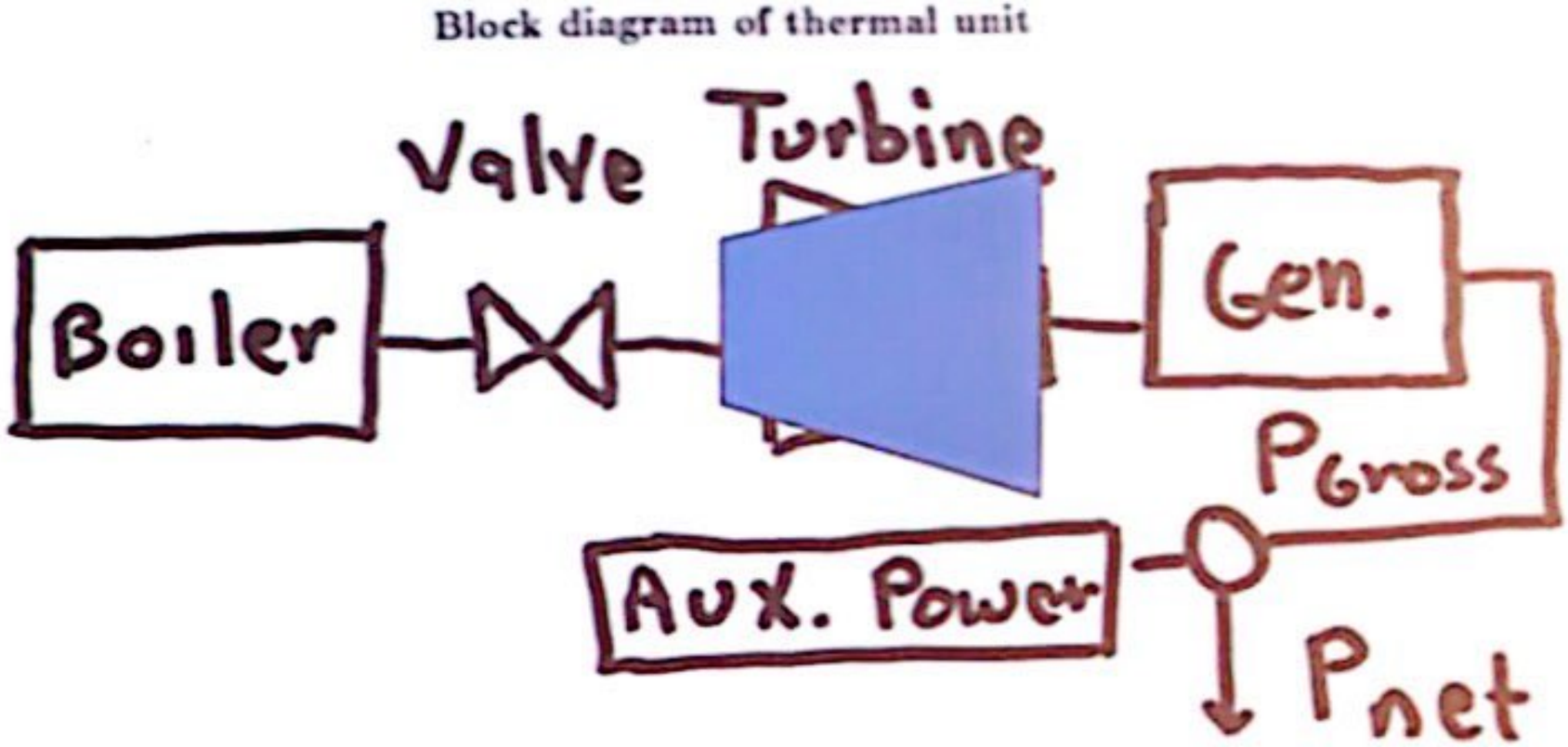
Generator types

- Traditionally utilities have had three broad groups of generators:
  - "Baseload" units: large coal/nuclear; almost always on at max.
  - "Midload," 'intermediate," or "cycling" units: smaller coal or gas that cycle on/off daily or weekly.
  - "Peaker" units: combustion turbines used only for several hours.



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Block Diagram of Thermal Unit



- To optimize generation costs we need to develop cost relationships between net power out and operating costs.
- Between 2-10% of power is used within the generating plant; this is known as the auxiliary power.

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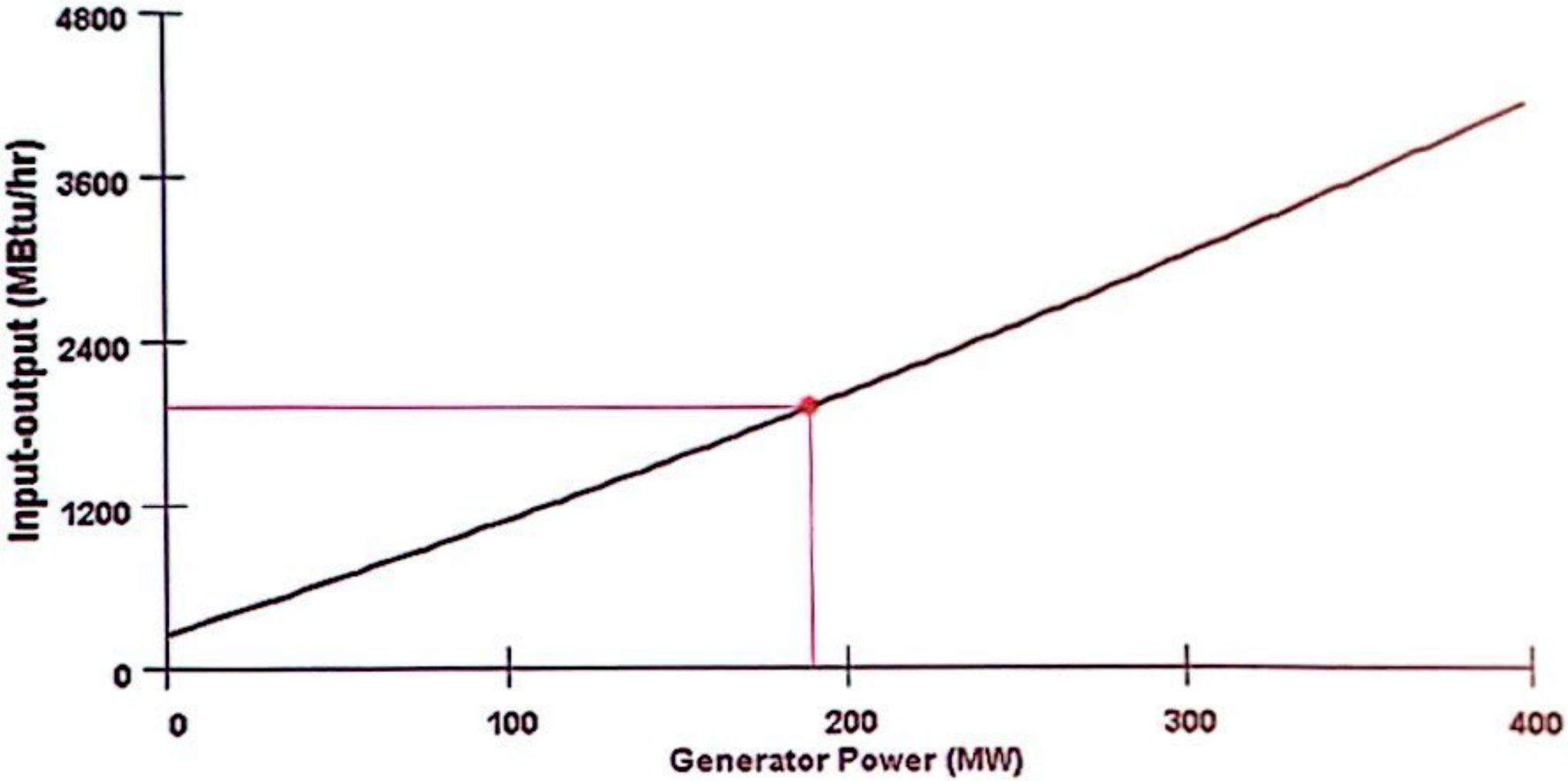
Thermal generator Cost Curves

- Thermal generator costs are typically represented by one or other of the following four curves
  - input/output (I/O) curve
  - fuel-cost curve
  - heat-rate curve
  - incremental cost curve
- For reference
  - 1 Btu (British thermal unit) = 1054 J
  - 1 MBtu = $1x10^6$ Btu
  - 1 MBtu = 0.29 MWh

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I/O Curve

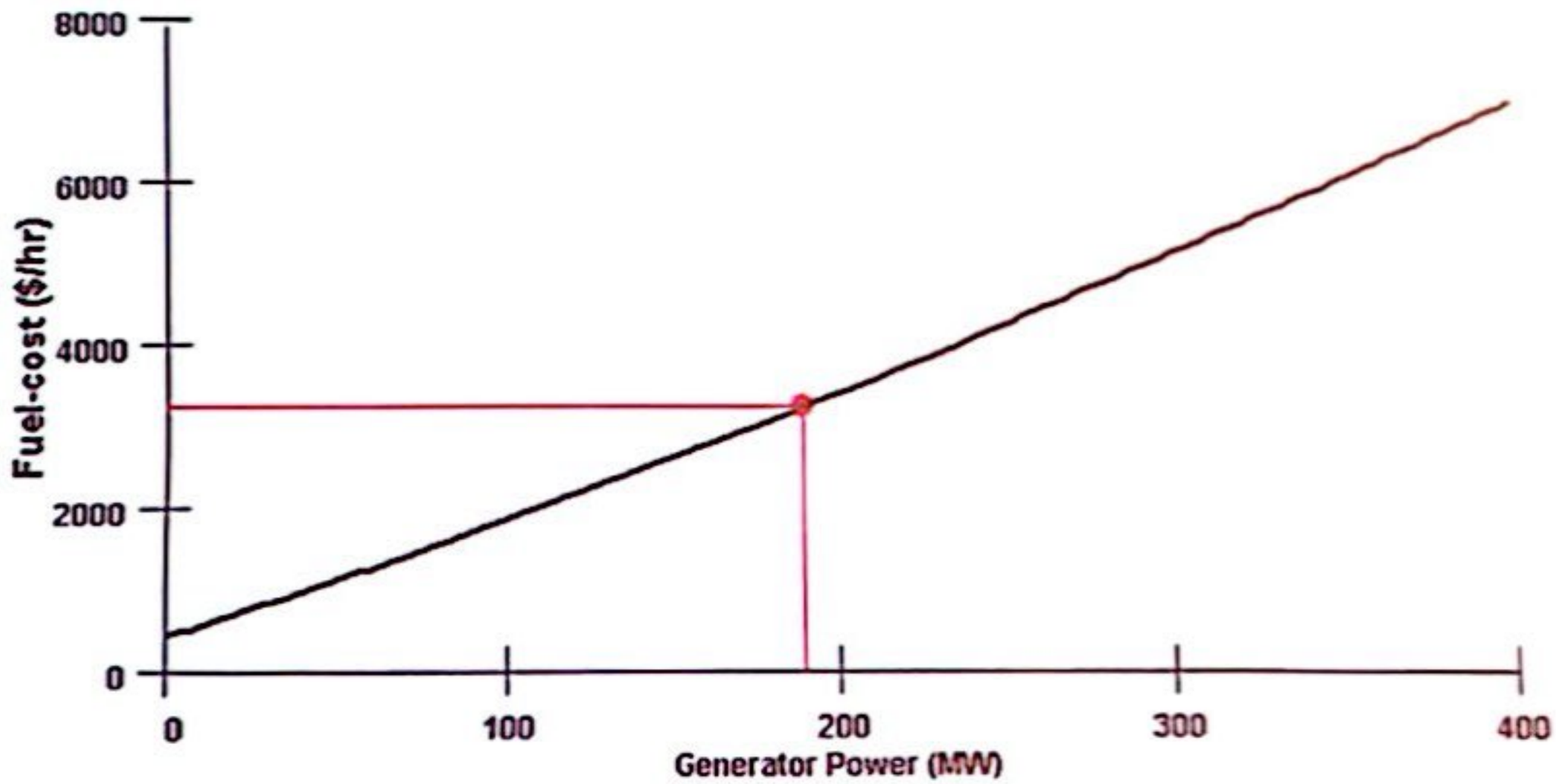
- The IO curve plots fuel input (in MBtu/hr) versus net MW output.



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Fuel-cost Curve

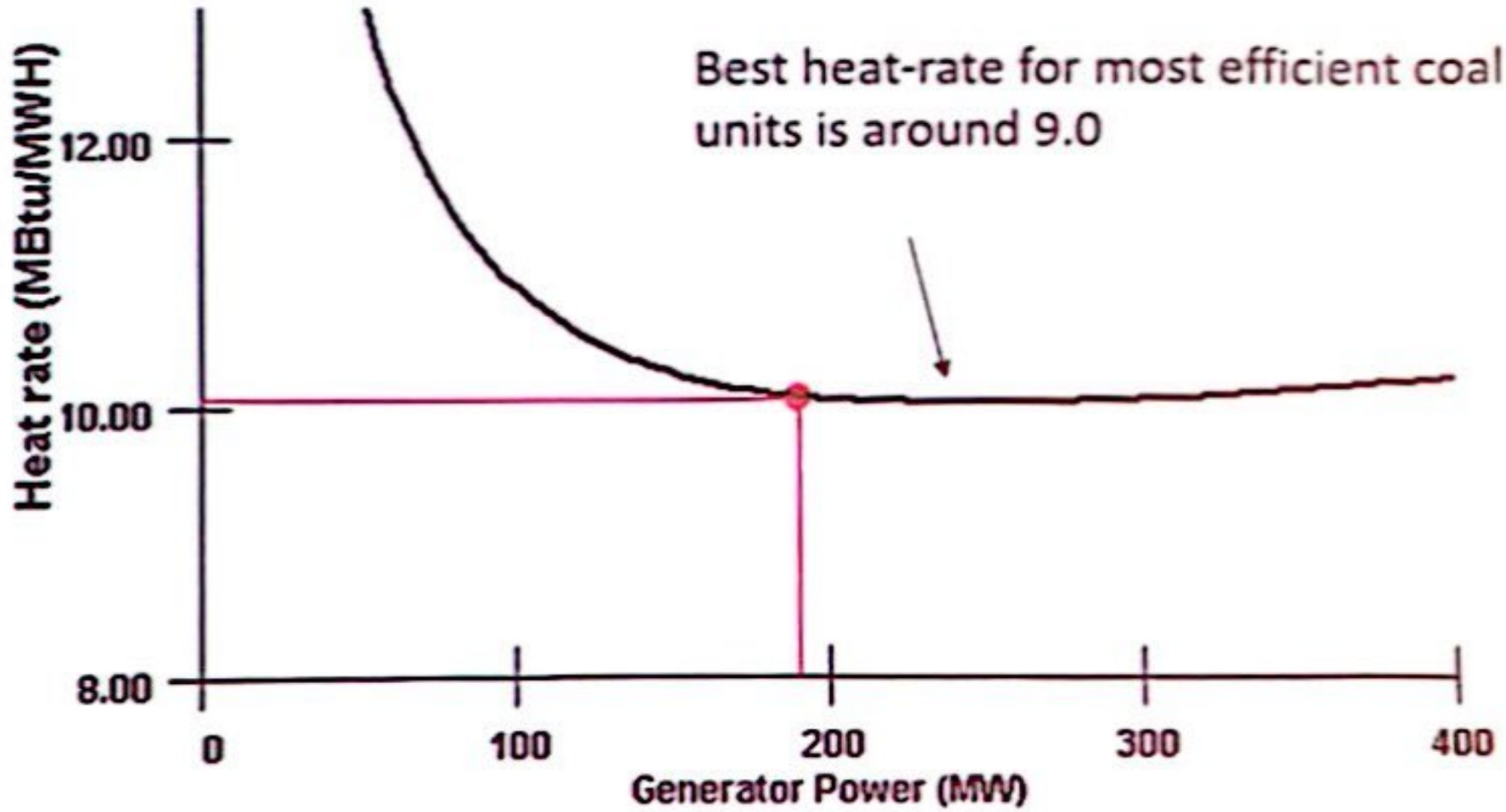
- The fuel-cost curve is the I/O curve multiplied by fuel cost.
- A typical cost for coal is $ 1.70/MBtu.



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Heat-rate Curve

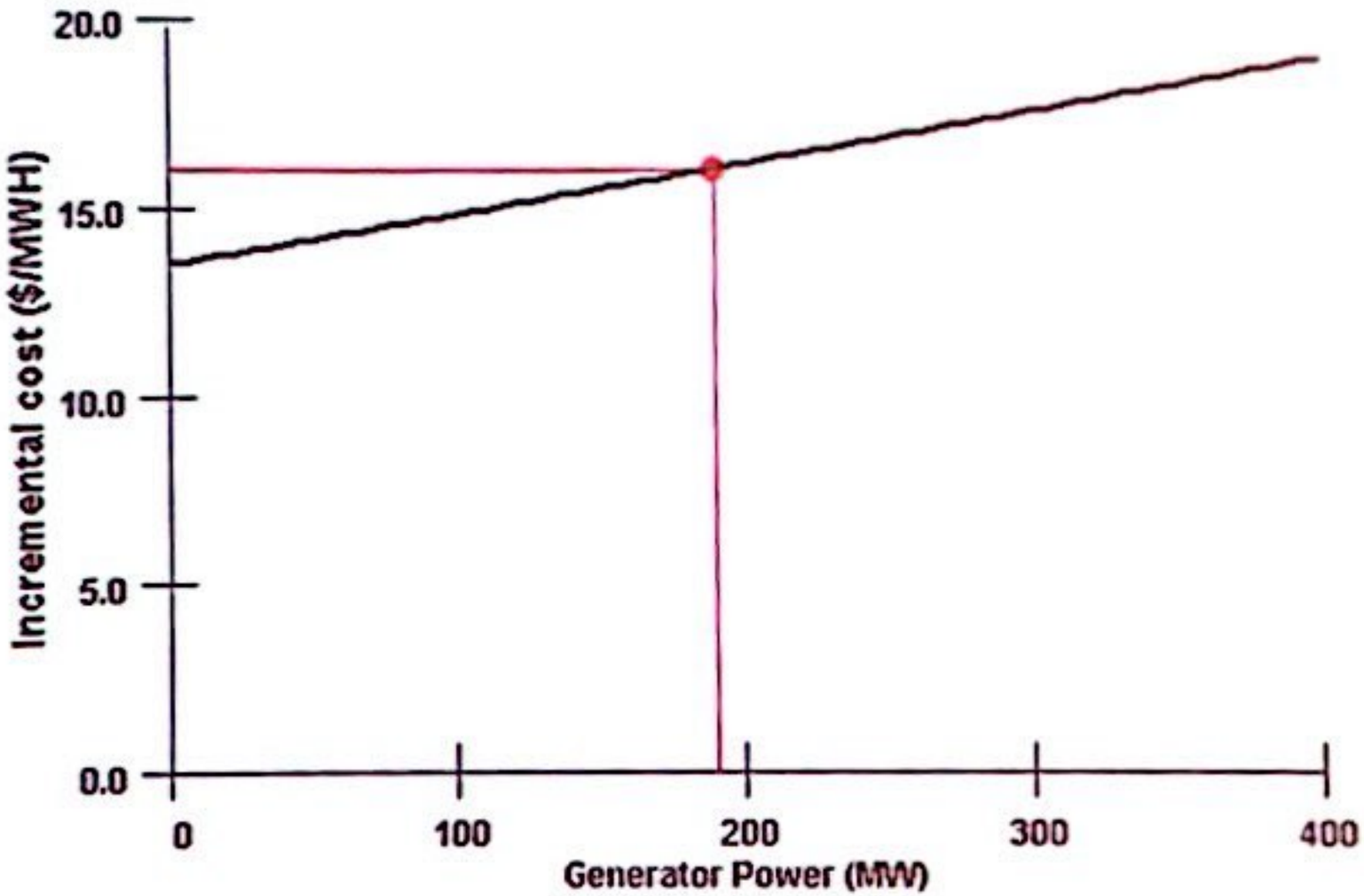
- Plots the average number of MBtu/hr of fuel input needed per MW of output.
- Heat-rate curve is the I/O curve divided by MW.



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Incremental (Marginal) cost Curve

- Plots the incremental $/MWh as a function of MW.
- Found by differentiating the cost curve.



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Coal Usage Example

- A 500 MW (net) generator is 35% efficient. It is being supplied with coal costing $1.70 per MBtu and with heat content 9000 Btu per pound. What is the coal usage in lbs/hr? What is the cost?

At 35% efficiency required fuel input per hour is

$$\frac{500\ \text{MWh}}{\text{hr} \times 0.35} = \frac{1428\ \text{MWh}}{\text{hr}} \times \frac{1\ \text{MBtu}}{0.29\ \text{MWh}} = \frac{4924\ \text{MBtu}}{\text{hr}}$$

$$\frac{4924\ \text{MBtu}}{\text{hr}} \times \frac{1\ \text{lb}}{0.009\text{MBtu}} = \frac{547{,}111\ \text{lbs}}{\text{hr}}$$

$$\text{Cost} = \frac{4924\ \text{MBtu}}{\text{hr}} \times \frac{\$1.70}{\text{MBtu}} = 8370.8\ \$/\text{hr or } \$16.74/\text{MWh}$$

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Wasting Coal Example

- Assume a 100W lamp is left on by mistake for 8 hours, and that the electricity is supplied by the previous coal plant and that transmission/distribution losses are 20%. How much coal has he/she wasted?

With 20% losses, a 100W load on for 8 hrs requires 1 kWh of energy. With 35% gen. efficiency this requires

$$\frac{1\ \text{kWh}}{0.35} \times \frac{1\ \text{MWh}}{1000\ \text{kWh}} \times \frac{1\ \text{MBtu}}{0.29\ \text{MWh}} \times \frac{1\ \text{lb}}{0.009\text{MBtu}} = 1.09\ \text{lb}$$

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Mathematical Formulation of Costs

- Generator cost curves are usually not smooth. However the curves can usually be adequately approximated using piece-wise smooth, functions.
- Two approximations predominate:
  - quadratic or cubic functions
  - piecewise linear functions
- We'll assume a quadratic approximation:

$$C_i(P_{Gi}) = \alpha_i + \beta_i P_{Gi} + \gamma_i P_{Gi}^2 \quad \$/\text{hr (fuel-cost)}$$

$$IC_i(P_{Gi}) = \frac{dC_i(P_{Gi})}{dP_{Gi}} = \beta_i + 2\gamma_i P_{Gi} \quad \$/\text{MWh}$$

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Incremental Cost Example

For a two generator system assume

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \text{ \$/hr}$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \text{ \$/hr}$$

Then

$$IC_1(P_{G1}) = \frac{dC_1(P_{G1})}{dP_{G1}} = 20 + 0.02P_{G1} \text{ \$/MWh}$$

$$IC_2(P_{G2}) = \frac{dC_2(P_{G2})}{dP_{G2}} = 15 + 0.06P_{G2} \text{ \$/MWh}$$

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Incremental Cost Example, cont'd

If $P_{G1} = 250$ MW and $P_{G2} = 150$ MW Then

$$C_1(250) = 1000 + 20 \times 250 + 0.01 \times 250^2 = \$\ 6625/\text{hr}$$

$$C_2(150) = 400 + 15 \times 150 + 0.03 \times 150^2 = \$6025/\text{hr}$$

Then

$$IC_1(250) = 20 + 0.02 \times 250 = \$\ 25/\text{MWh}$$

$$IC_2(150) = 15 + 0.06 \times 150 = \$\ 24/\text{MWh}$$

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Economic Dispatch Lagrangian

For the economic dispatch we have a minimization constrained with a single equality constraint

$$L(P_G, \lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D - \sum_{i=1}^{m} P_{Gi}) \quad \text{(no losses)}$$

The necessary conditions for a minimum are

$$\frac{\partial L(P_G, \lambda)}{\partial P_{Gi}} = \frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda = 0 \quad \text{(for i = 1 to m)}$$

$$P_D - \sum_{i=1}^{m} P_{Gi} = 0$$

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Economic Dispatch Example

What is economic dispatch for a two generator system $P_D = P_{G1} + P_{G2} = 500$ MW and

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \quad \$/hr$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \quad \$/hr$$

Using the Largrange multiplier method we know

$$\frac{dC_1(P_{G1})}{dP_{G1}} - \lambda = 20 + 0.02P_{G1} - \lambda = 0$$

$$\frac{dC_2(P_{G2})}{dP_{G2}} - \lambda = 15 + 0.06P_{G2} - \lambda = 0$$

$$500 - P_{G1} - P_{G2} = 0$$

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Economic Dispatch Example, cont'd

We therefore need to solve three linear equations

$$20 + 0.02P_{G1} - \lambda = 0$$

$$15 + 0.06P_{G2} - \lambda = 0$$

$$500 - P_{G1} - P_{G2} = 0$$

$$\begin{bmatrix} 0.02 & 0 & -1 \\ 0 & 0.06 & -1 \\ -1 & -1 & \end{bmatrix} \begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} -20 \\ -15 \\ -500 \end{bmatrix}$$

$$\begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} 312.5 \text{ MW} \\ 187.5 \text{ MW} \\ 26.2 \text{ \$/MWh} \end{bmatrix}$$

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Lambda-Iteration Algorithm

Pick λ^{L} and λ^{H} such that

$$\sum_{i=1}^{m} P_{Gi}(\lambda^{L}) - P_D < 0 \qquad \sum_{i=1}^{m} P_{Gi}(\lambda^{H}) - P_D > 0$$

While $\left|\lambda^{H} - \lambda^{L}\right| > \varepsilon$ Do

$$\lambda^{M} = (\lambda^{H} + \lambda^{L})/2$$

If $\sum_{i=1}^{m} P_{Gi}(\lambda^{M}) - P_D > 0$ Then $\lambda^{H} = \lambda^{M}$

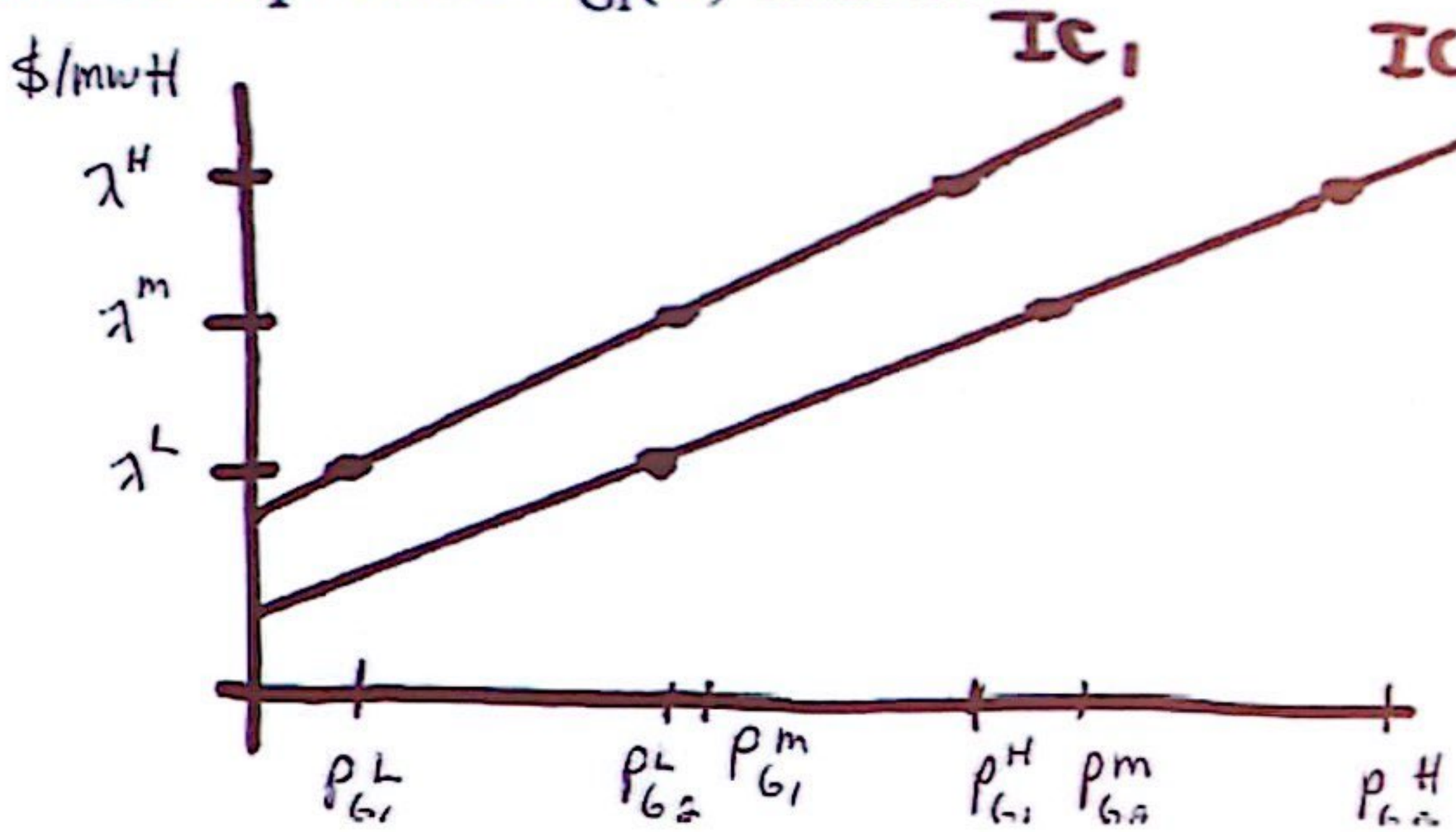
Else $\lambda^{L} = \lambda^{M}$

End While

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Lambda-Iteration: Graphical View

In the graph shown below for each value of lambda there is a unique P_{Gi} for each generator. This relationship is the $P_{Gi}(\lambda)$ function.



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Lambda-Iteration Example

Consider a three generator system with

$$IC_1(P_{G1}) = 15 + 0.02P_{G1} = \lambda \quad \text{\$/MWh}$$

$$IC_2(P_{G2}) = 20 + 0.01P_{G2} = \lambda \quad \text{\$/MWh}$$

$$IC_3(P_{G3}) = 18 + 0.025P_{G3} = \lambda \quad \text{\$/MWh}$$

and with constraint $P_{G1} + P_{G2} + P_{G3} = 1000\,\text{MW}$

Rewriting as a function of λ, $P_{Gi}(\lambda)$, we have

$$P_{G1}(\lambda) = \frac{\lambda - 15}{0.02} \qquad P_{G2}(\lambda) = \frac{\lambda - 20}{0.01}$$

$$P_{G3}(\lambda) = \frac{\lambda - 18}{0.025}$$

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Lambda-Iteration Example, cont'd

Pick convergence tolerance $\varepsilon = 0.05$ \$/MWh

Then iterate since $\left|\lambda^H - \lambda^L\right| > 0.05$

$$\lambda^M = (\lambda^H + \lambda^L)/2 = 25$$

Then since $\sum_{i=1}^{m} P_{Gi}(25) - 1000 = 280$ we set $\lambda^H = 25$

Since $|25 - 20| > 0.05$

$$\lambda^M = (25 + 20)/2 = 22.5$$

$$\sum_{i=1}^{m} P_{Gi}(22.5) - 1000 = -195 \text{ we set } \lambda^L = 22.5$$

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Lambda-Iteration Example, cont'd

Continue iterating until $\left|\lambda^{H} - \lambda^{L}\right| < 0.05$

The solution value of λ, λ^{*}, is 23.53 $/MWh

Once λ^{*} is known we can calculate the P_{Gi}

$$P_{G1}(23.5) = \frac{23.53 - 15}{0.02} = 426 \text{ MW}$$

$$P_{G2}(23.5) = \frac{23.53 - 20}{0.01} = 353 \text{ MW}$$

$$P_{G3}(23.5) = \frac{23.53 - 18}{0.025} = 221 \text{ MW}$$

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Generator MW Limits

- Generators have limits on the minimum and maximum amount of power they can produce
- Often times the minimum limit is not zero. This represents a limit on the generator's operation with the desired fuel type
- Because of varying system economics usually many generators in a system are operated at their maximum MW limits.

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Lambda-Iteration with Gen Limits

In the lambda-iteration method the limits are taken into account when calculating $P_{Gi}(\lambda)$:

if $P_{Gi}(\lambda) > P_{Gi.\max}$ then $P_{Gi}(\lambda) = P_{Gi.\max}$

if $P_{Gi}(\lambda) < P_{Gi.\min}$ then $P_{Gi}(\lambda) = P_{Gi.\min}$

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Lambda-Iteration Gen Limit Example

In the previous three generator example assume the same cost characteristics but also with limits

$0 \le P_{G1} \le 300$ MW $100 \le P_{G2} \le 500$ MW

$200 \le P_{G3} \le 600$ MW

With limits we get

$$\sum_{i=1}^{m} P_{Gi}(20) - 1000 = P_{G1}(20) + P_{G2}(20) + P_{G3}(20) - 1000$$

$$= 250 + 100 + 200 = -450 \text{ MW (compared to -670MW)}$$

$$\sum_{i=1}^{m} P_{Gi}(30) - 1000 = 300 + 500 + 480 - 1000 = 280 \text{ MW}$$

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Lambda-Iteration Limit Example,cont'd

Again we continue iterating until the convergence condition is satisfied. With limits the final solution of λ, is 24.43 $/MWh (compared to 23.53 $/MWh without limits). The presence of limits will always cause λ to either increase or remain the same.

Final solution is

$$P_{G1}(24.43) = 300 \text{ MW}$$
$$P_{G2}(24.43) = 443 \text{ MW}$$
$$P_{G3}(24.43) = 257 \text{ MW}$$

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Back of Envelope Values

- Often times incremental costs can be approximated by a constant value:
  - $/MWhr = fuelcost * heatrate + variable O&M
  - Typical heatrate for a coal plant is 10, modern combustion turbine is 10, combined cycle plant is 7 to 8, older combustion turbine 15.
  - Fuel costs ($/MBtu) are quite variable, with current values around 1.5 for coal, 4 for natural gas, 0.5 for nuclear, probably 10 for fuel oil.
  - Hydro, solar and wind costs tend to be quite low, but for this sources the fuel is free but limited

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Inclusion of Transmission Losses

- The losses on the transmission system are a function of the generation dispatch. In general, using generators closer to the load results in lower losses
- This impact on losses should be included when doing the economic dispatch
- Losses can be included by slightly rewriting the Lagrangian:

$$L(\mathbf{P_G}, \lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi})$$

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Impact of Transmission Losses

This small change then impacts the necessary conditions for an optimal economic dispatch

$$L(\mathbf{P_G},\lambda) = \sum_{i=1}^{m} C_i(P_{Gi}) + \lambda(P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi})$$

The necessary conditions for a minimum are now

$$\frac{\partial L(\mathbf{P_G},\lambda)}{\partial P_{Gi}} = \frac{dC_i(P_{Gi})}{dP_{Gi}} - \lambda(1 - \frac{\partial P_L(P_G)}{\partial P_{Gi}}) = 0$$

$$P_D + P_L(P_G) - \sum_{i=1}^{m} P_{Gi} = 0$$

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Impact of Transmission Losses

The condition for optimal dispatch with losses is then

$$L_1 IC_1(P_{G1}) = L_2 IC_2(P_{G2}) = L_m IC_m(P_{Gm}) = \lambda$$

Since $L_i = \dfrac{1}{\left(1 - \dfrac{\partial P_L(P_G)}{\partial P_{Gi}}\right)}$ if increasing P_{Gi} increases

the losses then $\dfrac{\partial P_L(P_G)}{\partial P_{Gi}} > 0 \rightarrow L_i > 1.0$

This makes generator i appear to be more expensive (i.e., it is penalized). Likewise $L_i < 1.0$ makes a generator appear less expensive.

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Calculation of Penalty Factors

Unfortunately, the analytic calculation of L_i is somewhat involved. The problem is a small change in the generation at P_{Gi} impacts the flows and hence the losses throughout the entire system. However, using a power flow you can approximate this function by making a small change to P_{Gi} and then seeing how the losses change:

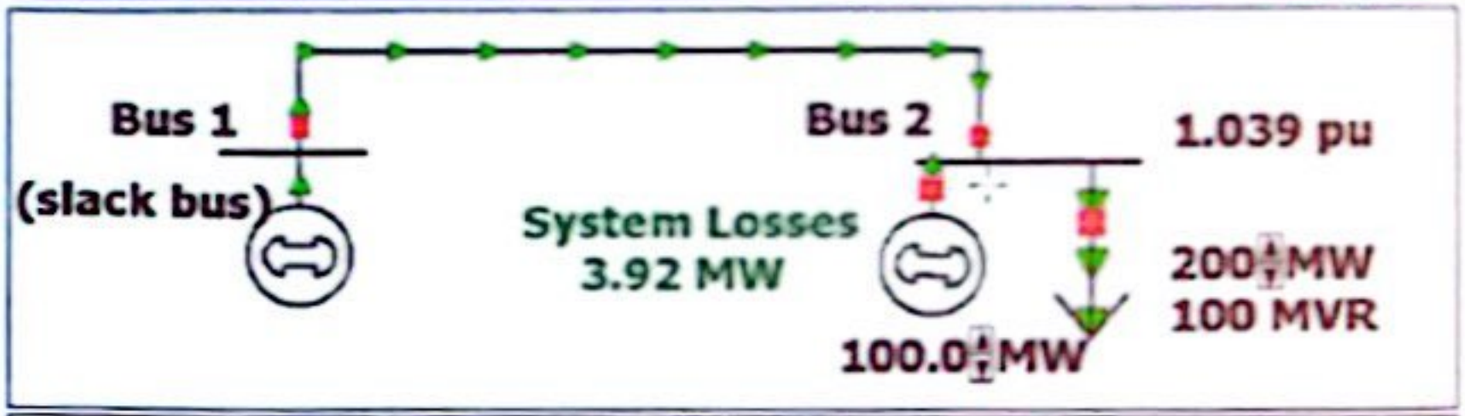
$$\frac{\partial P_L(P_G)}{\partial P_{Gi}} \approx \frac{\Delta P_L(P_G)}{\Delta P_{Gi}}$$

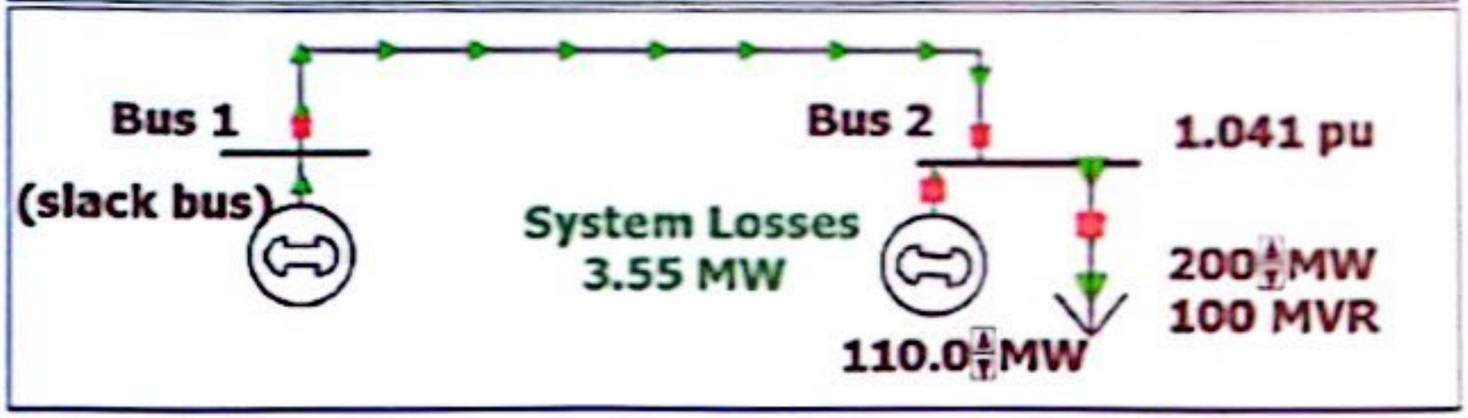
$$L_i \approx \frac{1}{1-\dfrac{\Delta P_L(P_G)}{\Delta P_{Gi}}}$$

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Two Bus Penalty Factor Example





$$\frac{\partial P_L(P_G)}{\partial P_{G2}} = -0.0387 \qquad \frac{\Delta P_L(P_G)}{\Delta P_{Gi}} = \frac{-0.37MW}{10MW} = -0.037$$

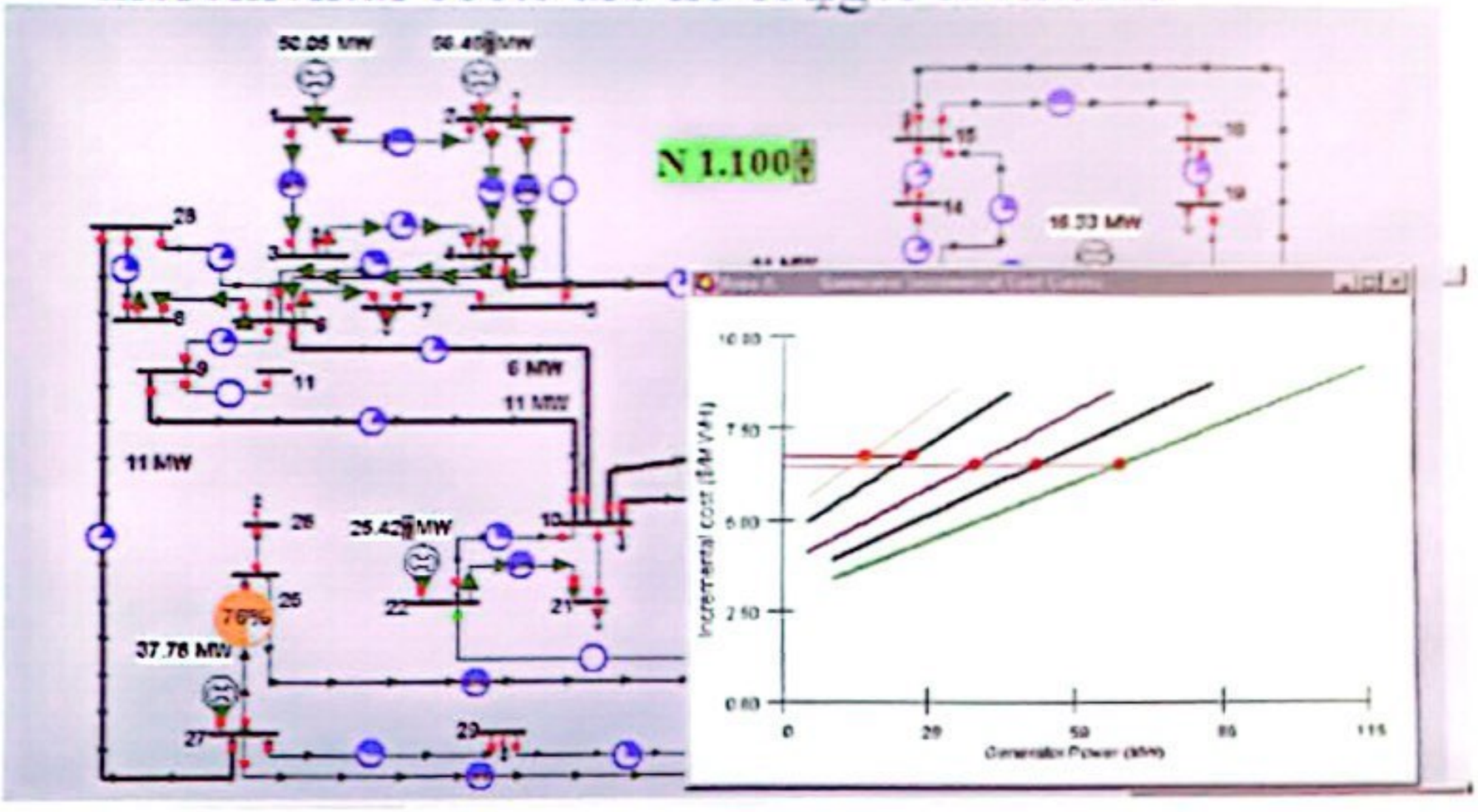
$$L_2 = 0.9627 \qquad L_2 \approx 0.9643$$

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Thirty Bus ED Example, cont

- Because of the penalty factors the generator incremental costs are no longer identical.



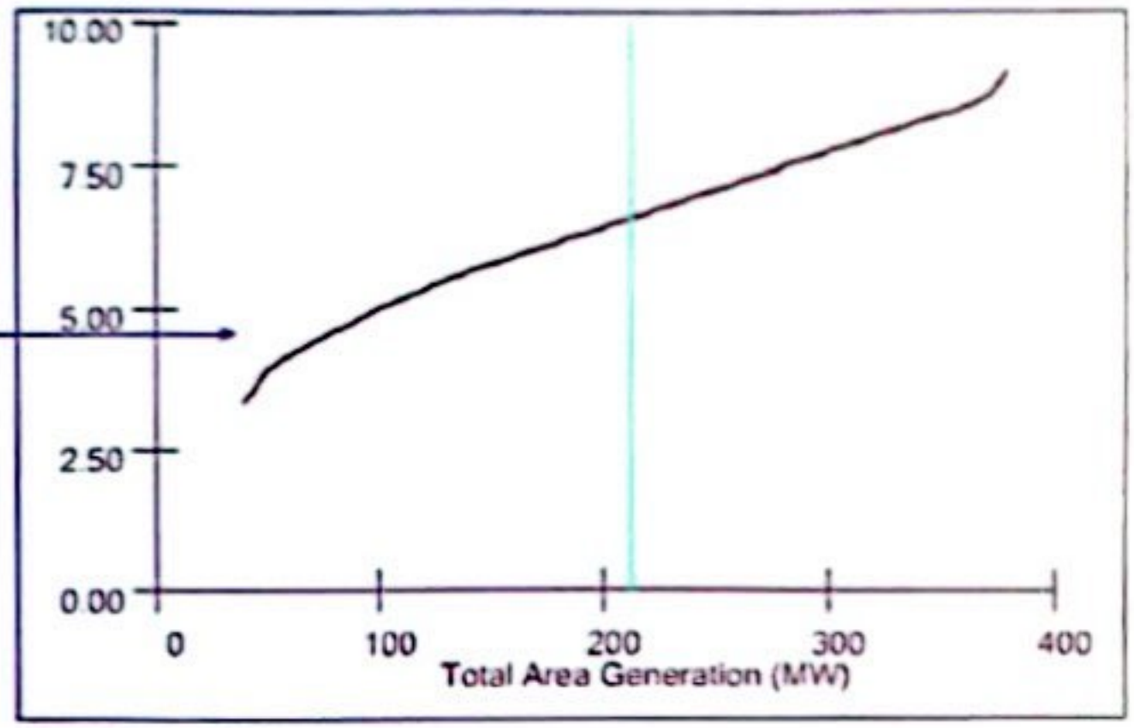
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Area Supply Curve

- The area supply curve shows the cost to produce the next MW of electricity, assuming area is economically dispatched

Supply curve for thirty bus system



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Araçlar İmzala Yorum

Economic Dispatch - Summary

- Economic dispatch determines the best way to minimize the current generator operating costs
- The lambda-iteration method is a good approach for solving the economic dispatch problem
  - generator limits are easily handled
  - penalty factors are used to consider the impact of losses
- Economic dispatch is not concerned with determining which units to turn on/off (this is the unit commitment problem)
- Economic dispatch ignores the transmission system limitations

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Araçlar İmzala Yorum

- Example
  - consider the cost for operating three generation units

Unit 1:	$F_1(P_1) = 561 + 7.92\,P_1 + 0.001562\,P_1^2$	$150 \le P_1 \le 600$
Unit 2:	$F_2(P_2) = 310 + 7.85\,P_2 + 0.00194\,P_2^2$	$100 \le P_2 \le 400$
Unit 3:	$F_3(P_3) = 93.6 + 9.56\,P_3 + 0.005784\,P_3^2$	$50 \le P_3 \le 200$

  - what combination of units is best to supply a 550 MW load?

Unit 1	Unit 2	Unit 3	Max Gen	Min Gen	P_1	P_2	P_3	F_1	F_2	F_3	F_{total}
Off	Off	Off	0	0	Infeasible						
Off	Off	On	200	50	Infeasible						
Off	On	Off	400	100	Infeasible						
Off	On	On	600	150	0	400	150	0	3760	1658	5418
On	Off	Off	600	150	550	0	0	5389	0	0	5389
On	Off	On	800	200	500	0	50	4911	0	586	5497
On	On	Off	1000	250	295	255	0	3030	2440	0	5471
On	On	On	1200	300	267	233	50	2787	2244	586	5617

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